**A**

**Mini Project Report**

**on**

# Public Complaint Registration App

Submitted in partial fulfillment of the requirements for the

degree

**Second Year Engineering – Computer Science Engineering (Data Science)**

by


| | |
|---|---|
| **Bhumi Koli** | **25207010** |
| **Kashish Jain** | **24107031** |
| **Namrata Dhas** | **25207012** |
| **Sanika Pawar** | **25207015** |

**Under the guidance of**

**Ms. Sarala Mary**

**DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (DATA SCIENCE)**

A.P. SHAH INSTITUTE OF TECHNOLOGY

G.B. Road, Kasarvadavali, Thane (W)-400615

UNIVERSITY OF MUMBAI


**Academic year: 2025-26**

# CERTIFICATE

This to certify that the Mini Project report on **Public Complaint Registration App** has been submitted by **Bhumi Koli (25207010)**, **Kashish Jain (24107031), Namrata Dhas (25207012)** and **Sanika Pawar (25207015)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

**Ms. Sarala Mary**
**Guide**

**Dr. Pravin Adivarekar**
**HOD, CSE(Data Science)**

**Dr. Uttam D. Kolekar**
**Principal**

**External Examiner:**
**1.**

**Internal Examiner:**
**1.**

**Place:** A. P. Shah Institute of Technology, Thane
**Date:**

# ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Ms. Sarala Mary.** Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Mrs.Hardiki Patil and Mrs.Dipali Gat** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

# TABLE OF CONTENTS

1. Introduction……..1

   1.1.Purpose.....1

   1.2.Problem Statement……...2

   1.3.Objectives..... 3

   1.4.Scope.....…...3

2. Proposed System…….....4

   2.1.Features and Functionality……...5

3. Project Outcomes.....…....6

4. Software Requirements.....8

5. Project Design……..…….....10

6. Project Scheduling.....12

7. Results……13

8. Conclusion…….....18

References

# Chapter 1

# Introduction

The **CivicVoice** project is a comprehensive mobile-based platform developed to bridge the gap between citizens and municipal authorities regarding local public issues. In many cities, people find it extremely difficult to report common problems like broken roads, garbage piles, water leakages, or malfunctioning streetlights because there is no simple or direct way to reach the right officials. Even when a complaint is successfully made through traditional methods, it is often delayed, ignored, or lost because there is no centralized system to track the progress of the work. This lack of communication leads to frustration among residents and allows public infrastructure to stay in poor condition for long periods.

To solve these problems, we have developed a robust mobile application using **Flutter** for the user interface and **Firebase** for the backend database. This application allows citizens to register their complaints instantly and accurately by uploading a photo of the issue and using the **Google Maps API** to automatically capture the exact GPS location. By providing authorities with both photographic evidence and a precise location, the system eliminates any confusion about where the problem is or how serious it has become. Furthermore, the app includes a real-time tracking system where users can see if their complaint is "Pending," "In Progress," or "Resolved," ensuring that the government remains accountable and that work is actually completed.

## 1.1. Purpose:

The primary purpose of this document is to outline the development and implementation of the **Civic Voice** mobile application, which serves as a digital bridge between citizens and municipal authorities. The document defines a structured approach to solving public infrastructure problems such as road damage, garbage, and water leakage—by moving away from traditional, inefficient reporting methods toward a transparent, real-time digital system.

The target audience for this document and the associated project includes:

- **Citizens**: Individuals who require a user-friendly mobile tool to report local grievances with photo and location support while tracking the progress of their complaints in real time.
- **Municipal Authorities**: Government officials and administrators who need an organized system to view, track, and manage public complaints more effectively to ensure faster resolutions.
- **Volunteers:** Community groups and non-governmental organizations that can use the platform to discover specific local maintenance tasks, such as cleaning drives, to assist in improving their neighbourhoods.
- **System Developers and Designers**: The technical team responsible for maintaining the technology stack, which involves **Flutter** for the frontend, **Firebase** for the backend, and various **APIs** for camera and mapping functionality.

## 1.2. Problem Statement:

a) **Difficulty in Reporting:** Citizens find it very hard to report local issues like broken roads, garbage, or water leaks because there is no easy way to contact the right people.

b) **No Way to Track Progress:** There is no proper system for people to see if their complaint is being worked on, which leads to long delays and confusion.

c) **Location Confusion:** Without GPS, authorities often have trouble finding the exact spot where the problem is located, making it harder to fix.

d) **Too Much Paperwork:** Most complaints are currently handled using slow, manual paperwork, which makes the whole process inefficient for the government.

e) **Lack of Proof:** There is often no clear evidence to show whether a problem was actually solved or if the work was done correctly.

## 1.3. Objectives:

a) **Smart Complaint Reporting with Photos and GPS:** Users can easily report issues by uploading real-time photos and using GPS to share the exact location, helping authorities find and understand the problem quickly.

b) **Live Status Tracking and Progress Management:** The system provides a clear way for citizens to track their complaints through stages like "Pending," "In Progress," and "Resolved" so they stay informed about the work.

c) **Community Voting for Faster Solutions:** A public voting system lets citizens support the most urgent issues, which helps authorities prioritize and fix high-demand problems first.

d) **Proof of Work with Photo Verification:** To ensure the job is done correctly, the system requires "Before" and "After" photos as digital evidence that the reported issue has been fully resolved.

e) **Digital Admin Dashboard:** An online dashboard for authorities replaces slow paperwork with a fast, system for viewing, updating, and managing all public complaints.

## 1.4. Scope:

a) **City-Wide Implementation:** The application can be used in cities and municipal offices to report and manage local issues like road damage, garbage, and water leaks.

b) **Mobile Accessibility:** It provides a simple way for citizens to send detailed complaints through a mobile app, including photos and precise locations.

c) **Authority Support:** The platform helps government authorities track and solve public problems much faster using a digital system.

d) **Full System Management:** The scope includes a mobile app for reporting and a backend system to store and update the status of every complaint

# Chapter 2

# Proposed System

The **Proposed System** is a modern mobile platform that makes it easy for citizens to report local problems and for authorities to fix them quickly. It replaces old, slow paperwork with a fast digital system that uses photos and GPS to show exactly what and where the issues are.

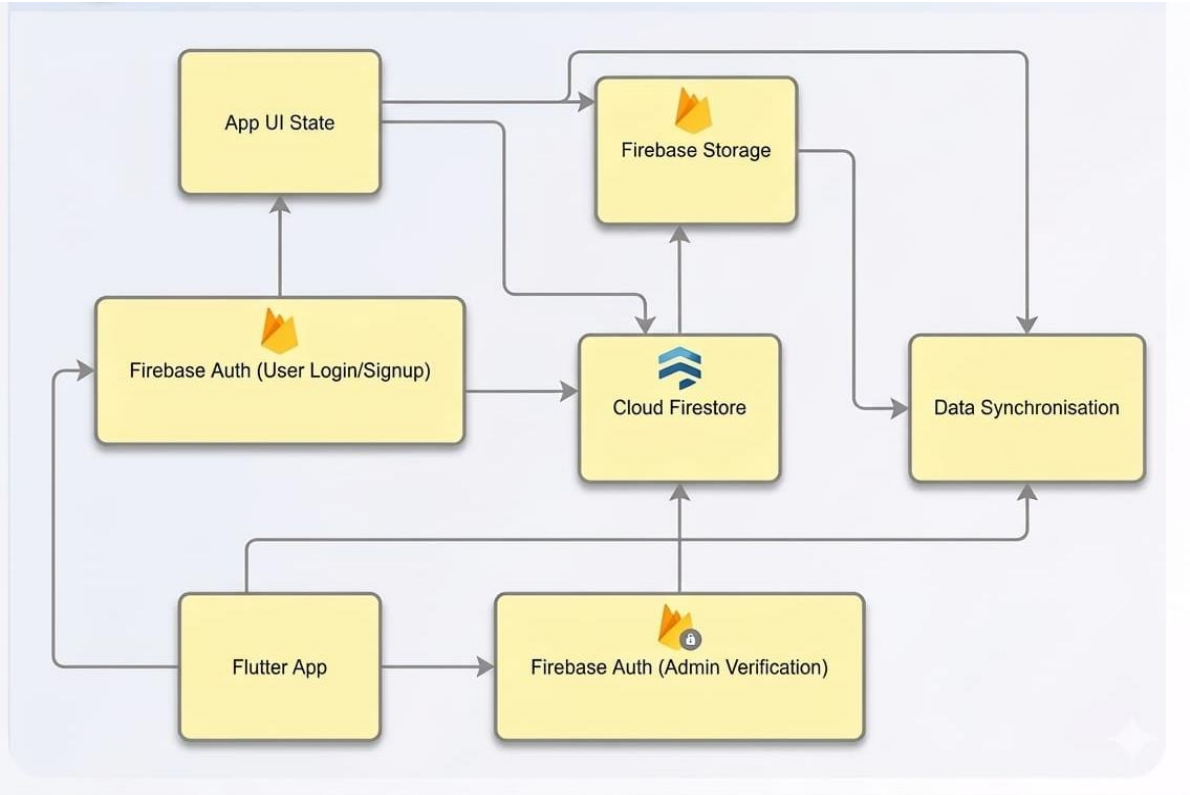


**Fig 2.1: Block Diagram**

The above **Fig 2.1: block diagram** shows how the different parts of the app work together to handle a complaint:

- **Flutter App:** This is the actual mobile application that users and admins open on their phones to report issues or manage tasks.
- **Firebase Auth:** This part handles the security; it makes sure users are logged in and verifies the admin before they can update any data.
- **App UI State:** This manages what you see on your screen, like the list of complaints or the "Pending" and "Resolved" buttons.

- **Cloud Firestore:** This is the main database that stores all the text information, such as the complaint description and the location coordinates.
- **Firebase Storage:** This is a special digital folder that stores all the photos users take and upload as proof of the problem.
- **Data Synchronisation:** This ensures that when an admin fixes a problem and updates the status, the user sees that change instantly on their own app.

## 2.1 Features and Functionality:

a) **User Complaint Submission**: Citizens can easily register a complaint by entering a text description and selecting a specific category for the issue.

b) **Multimedia Evidence Support**: The app uses the **Flutter Camera API** so users can take and upload real-time photos of the problem as proof.

c) **Location Detection**: By integrating the **Google Maps API**, the system captures the exact GPS location of the issue to help authorities find it easily.

d) **Real-Time Status Tracking**: Users can monitor the progress of their complaints through clear status updates like **Pending**, **In Progress**, or **Resolved**.

e) **Secure User Authentication**: The system uses **Firebase Auth** to provide a secure login and signup process for both citizens and administrative staff.

f) **Admin Management Dashboard**: Authorities have a dedicated panel to view all incoming complaints and update their status after the work is done.

g) **Cloud Data Storage**: All complaint details, images, and locations are stored and synchronized securely using **Firebase** and **Cloud Firestore**

# Chapter 3

# Project Outcomes

a) **Accurate Problem Identification with Visual and Spatial Proof: -**

The main result is that the government receives a complete "digital package" for every issue reported. Instead of just a text message saying "there is trash," they get a clear photo and a precise map pin. This outcome means workers don't waste time driving around looking for the right spot or bringing the wrong tools. They know exactly what the problem looks like and where it is before they even leave the office, which makes the entire repair process much faster and more organized.

b) **Open and Honest Communication for Every Citizen: -**

This outcome removes the "mystery" from government work and reduces the gap between the public and authorities. In the past, people felt ignored because they didn't get updates and had to make multiple phone calls to check on a complaint. Now, the app gives a clear, live timeline for every request. When a user sees their status change from "Pending" to "In Progress," it provides peace of mind that a team is actually on the way, creating a much stronger bond of trust between the residents and city workers.

c) **Fair and Smart Decision Making through Community Support: -**

The voting system creates an outcome where the community directly influences which problems get solved first. This makes the system fair because the most popular or urgent complaints naturally rise to the top of the admin's list. It helps the government focus their limited budget, equipment, and manpower on the problems that are bothering the largest number of residents first. This ensures that major issues, like a broken water pipe affecting many families, are not delayed by minor, individual requests.

d) **Guaranteed High-Quality Work through Photo Evidence: -**

This result ensures that work is not just marked as done, but is actually finished to a high standard. By comparing the "Before" photo submitted by the citizen and the "After" photo uploaded by the worker, the system creates a permanent, digital record of the repair. This outcome stops anyone from closing a file without doing the work, ensuring that public money is being spent on real results. It also gives the citizen a sense of satisfaction, as they receive visual confirmation that their specific problem has been solved.

e) **A Faster, Paperless, and Modernized Government System: -**

The final outcome is the complete removal of slow, messy paperwork and physical files that often get lost. By using a Digital Admin Dashboard, the city administration can track hundreds of complaints at once from a single computer screen. This results in much faster response times and allows the government to see city-wide patterns like which neighbourhoods have the most frequent drainage issues so they can plan long-term fixes instead of just quick repairs, making the entire city administration more efficient.

# Chapter 4

# Software Requirements

Here are the project tools and requirements explained in very simple language, point-by-point:

### 4.1 Development Tools (The Workspace)

- **Flutter SDK:** The main toolkit used to design and build the app for both Android and iPhones.
- **Dart:** The specific coding language used to write all the app's features.
- **IntelliJ idea:** The software programs on the computer where the code is actually written.
- **Firebase Console:** An online dashboard used to set up the app’s "brain" (the backend and database).
- **Git & GitHub:** Tools used to save different versions of the code so no work is ever lost.

### 4.2 App Parts (The Building Blocks)

- **firebase_core:** The "bridge" that connects the app to the online database.
- **firebase_auth:** The security guard that handles user logins and keeps accounts safe.
- **cloud_firestore:** A digital notebook that stores all the text, like complaint details and user names.
- **firebase_storage:** A digital locker where all the photos taken by users are kept safely.
- **google_maps_flutter:** The tool that puts a real map inside the app.
- **geolocator:** The part that finds the user's exact location (GPS) automatically.
- **camera / image_picker:** The feature that lets the app open the phone's camera to take pictures.
- **provider:** A system that makes sure the app updates instantly when something changes.

### 4.3 Hardware Needs (The Equipment)

- **Smartphones:** At least one Android or iPhone to test if the app works correctly in real life.
- **Testing Software:** Computer programs (emulators) that act like fake phones for quick testing.
- **Laptop/PC:** A computer with enough power (8GB RAM) to run the coding tools smoothly.
- **Internet:** A stable connection to send and receive data from the cloud and maps.

### 4.4 Quality Requirements (The Standards)

- **Speed:** The app should open very fast (under 3 seconds) and never feel laggy or slow.
- **Safety:** Only the right people (admins) can change a complaint's status to "Fixed."
- **Easy to Use:** The buttons and menus are made to be very simple so anyone can use them.
- **Offline View:** Users can still look at their old complaints even if they don't have internet at that moment.

# Chapter 5

# Project Design

The system architecture follows a **client-server model** with **Firebase** as the backend. The key components are the **Flutter mobile app**, **Firebase cloud services**, and the **Google Maps API** for location tracking.

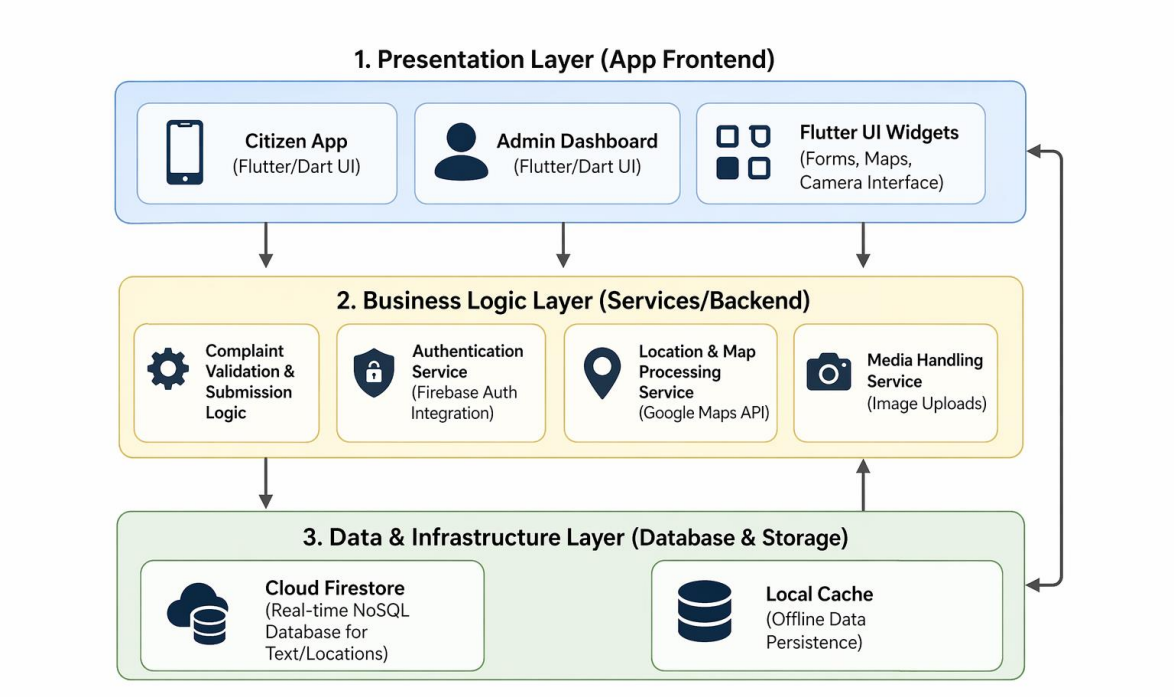


**Fig 5.1: Layered Architecture of Application**

The above **Fig 5.1: Layered Architecture of Application** shows how the app is built in three main levels (layers) that talk to each other:

- **1. Presentation Layer (The Frontend):**
  - **Citizen App:** The part where people report problems and check their status.
  - **Admin Dashboard:** The part where officials log in to see and solve complaints.
  - **Flutter UI:** The buttons, forms, and maps that make the app easy to use.
- **2. Business Logic Layer (The Backend):**
  - **Login Service:** Keeps accounts safe using Firebase.
  - **Location Service:** Uses Google Maps to find exactly where the problem is.

    - **Media Service:** Handles the photos taken by the camera.
- **3. Data Layer:**
    - **Cloud Firestore:** A digital notebook that saves all the names, dates, and complaint details.
    - **Local Cache:** Saves a copy on the phone so the app works even when the internet is slow.

## System Overview

- **Reporting Problems:** A citizen opens the app, takes a photo, and hits "Submit." The app automatically grabs their GPS location so the authorities know exactly where to go.
- **Secure Access:** The app uses Firebase to make sure only real users can post complaints and only real admins can change the status of a complaint.
- **Real-Time Updates:** When an admin changes a complaint from **"Pending"** to **"Resolved,"** the citizen sees that update on their phone instantly without having to refresh the page.
- **Map Integration:** The Admin can see all complaints as "pins" on a map. This helps them see which areas have the most problems (like a lot of garbage or broken lights in one colony).
- **Offline Support:** If a user is in an area with no network, they can still open the app and see their past complaints because the data is saved locally on their device.

# Chapter 6

# Project Scheduling

## Gantt Chart Description

The Below **Fig 6.1: Gantt Chart for Project Scheduling** project development is scheduled from January to March covering all phases from requirement analysis to final documentation.

**January:** Requirement analysis and system design

**Feburary:** Core development and testing

**March:** Database design and initial development & Final implementation and documentation

**Project Name:** Public Complaint Registration App
**Project Duration:** Jan – March

| WBS | ACTIVITY | ASSIGNED TO | % DONE | START | DURATION |
|---|---|---|---|---|---|
| PHASE 1: Project Planning & Requirement Analysis | | | | | |
| 1.1 | Topic Finalization & Group Formation | A,B,C | 100% | 10-Jan | 1 Week |
| 1.2 | Requirement Gathering | A,B | 100% | 22-Jan | 1 Week |
| 1.3 | System Analysis & Feasibility Study | B,C | 100% | 30-Jan | 1 Week |
| PHASE 2: System Design | | | | | |
| 2.1 | UI/UX Design (App Screens) | A | 1-Feb | 10-Feb | 1.5 Weeks |
| 2.2 | Database Design | B | 5-Feb | 12-Feb | 1 Week |
| 2.3 | System Architecture Design | C | 8-Feb | 15-Feb | 1 Week |
| PHASE 3: Development | | | | | |
| 3.1 | Backend Development | B,& C | 16-Feb | 1-Mar | 2 Weeks |
| 3.2 | Frontend Development | A | 20-Feb | 7-Mar | 2.5 Weeks |
| 3.3 | Testing & Bug Fixing | A,B,C | 5-Mar | 21-Mar | 2.5 Weeks |

**Figure 6.1: Gantt Chart for Project Scheduling**

# Chapter 7

# Results

The results of the CivicVoice system using different screenshots. It explains how users can file and track complaints and how admins manage them easily.

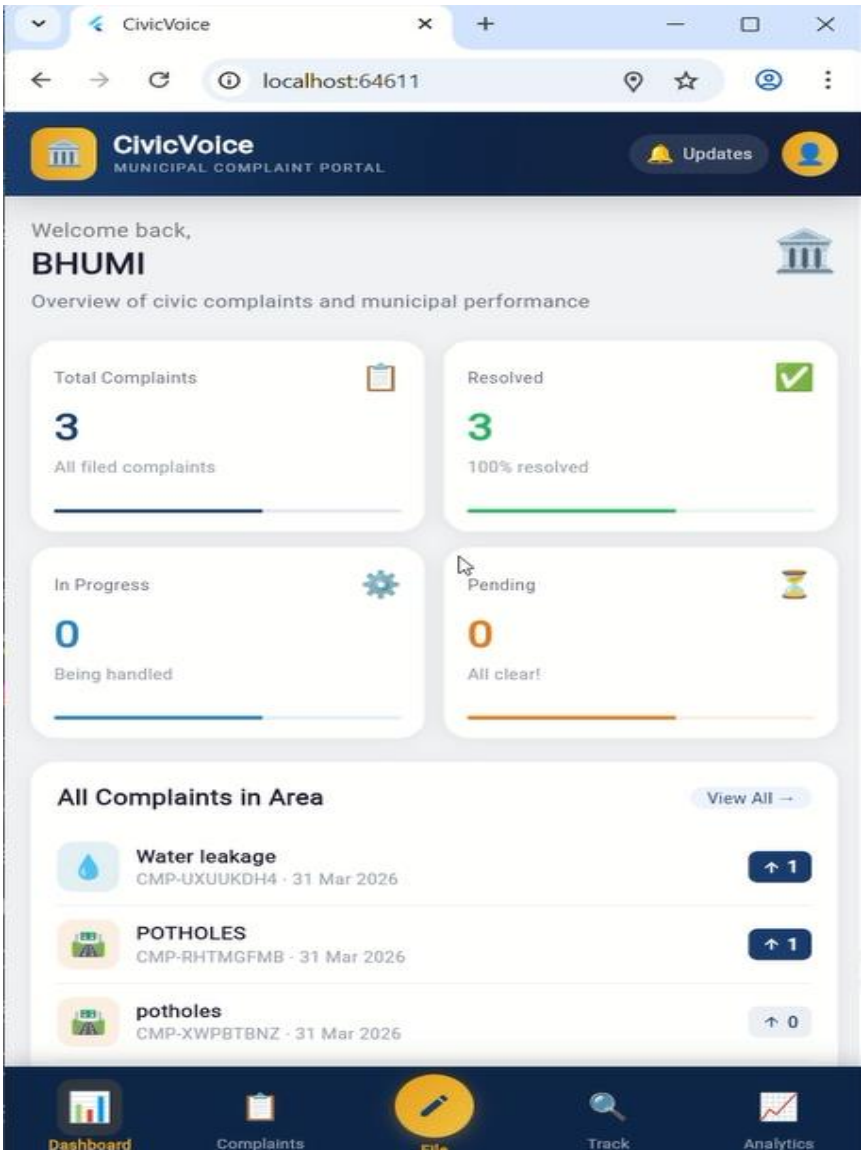


**Fig 7.1: User Dashboard**

This Above **fig 7.1: User Dashboard** shows a **CivicVoice municipal complaint dashboard** for a user. It summarizes complaint status total, resolved, in progress, and pending along with a list of reported issues like water leakage and potholes. The interface helps users track and manage civic complaints easily.

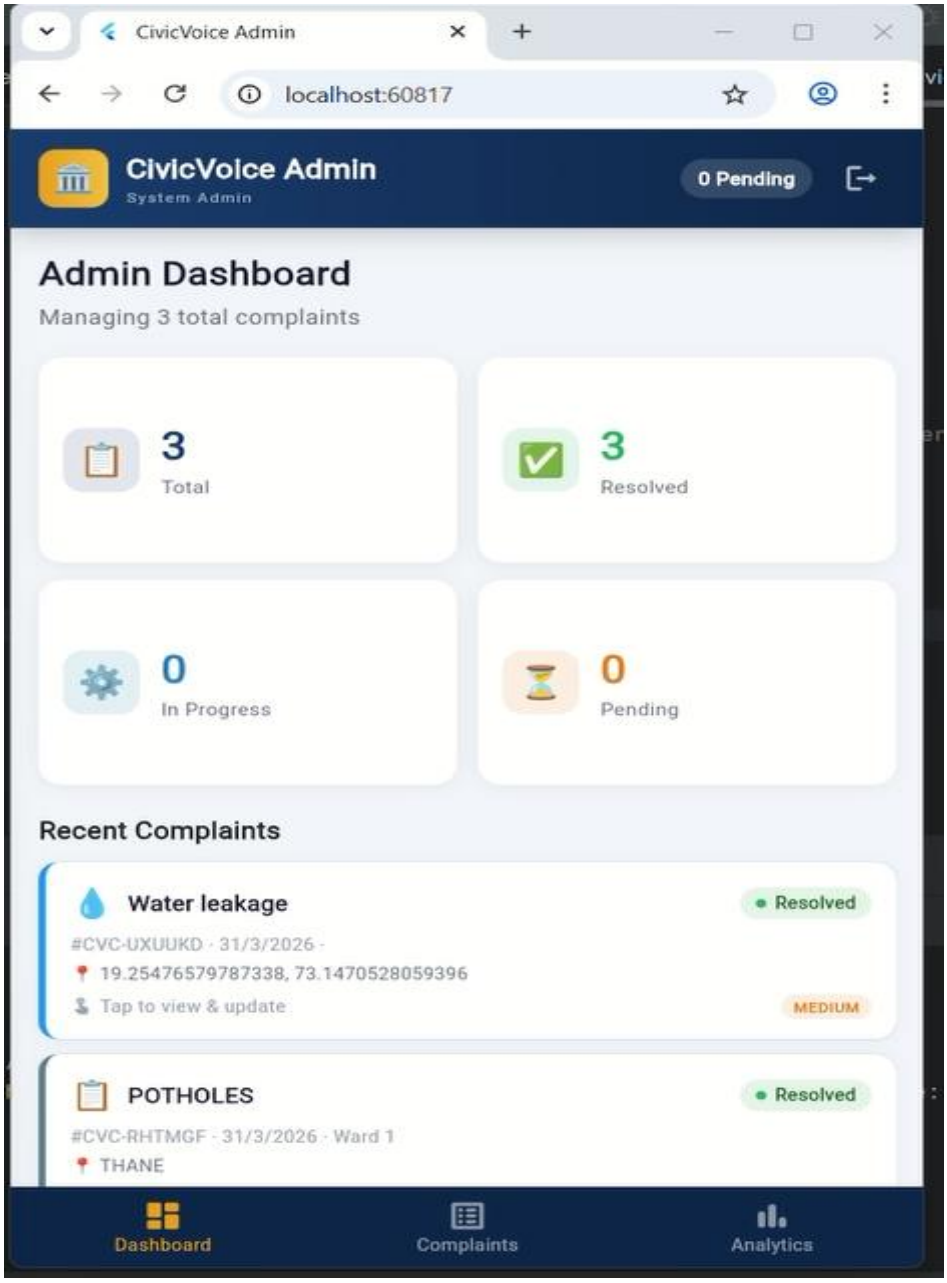


**Fig 7.2: Admin Dashboard**

**Fig 7.2: Admin Dashboard** shows the CivicVoice system admin panel used to manage municipal complaints. It provides a summary of total, resolved, in-progress, and pending complaints along with recent complaint details like water leakage and potholes. The interface helps admins monitor, update, and efficiently handle complaint records.

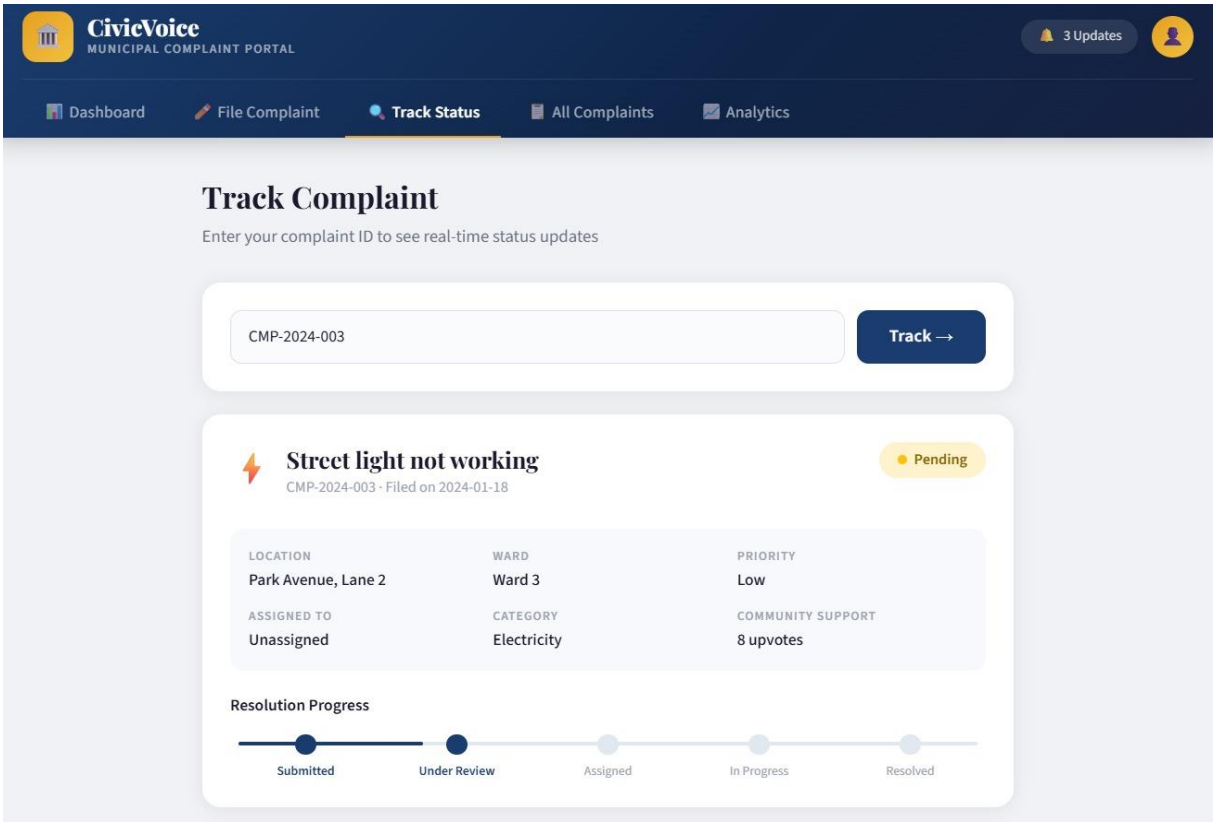


**Fig 7.3: Live Status tracking of complaint**

**Fig 7.3: Live Status Tracking of Complaint** shows the CivicVoice complaint tracking interface where users can enter a complaint ID to view real-time updates. It displays detailed information such as location, category, priority, and resolution progress from submission to completion. The interface helps users monitor the status and progress of their complaints efficiently.

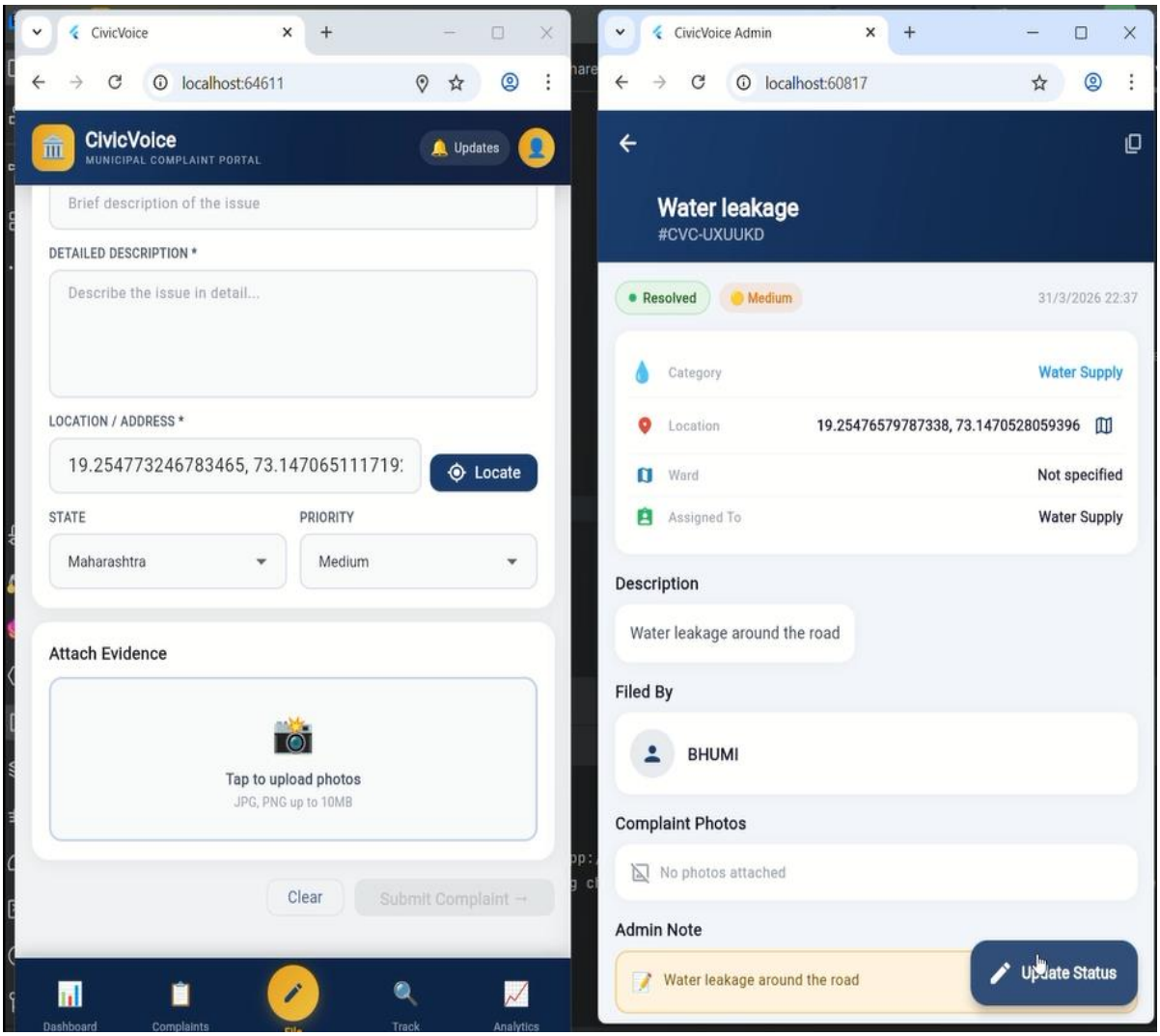


**Fig 7.4: File a complaint**

**Fig 7.4: File a Complaint** shows how a user can report an issue by filling details like description, location, and uploading photos. It also shows the admin side where the complaint details and status can be viewed and updated easily.

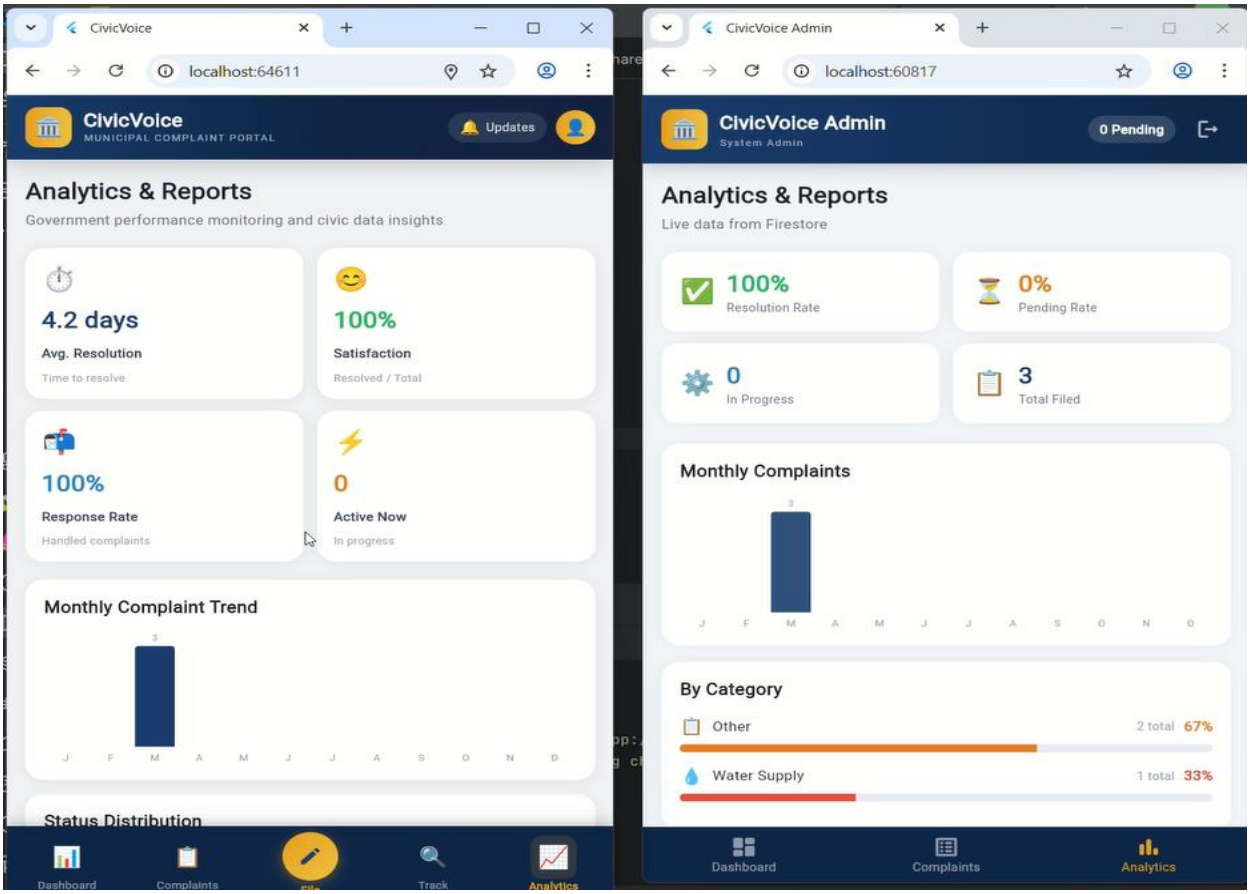


**Fig 7.5: Analytics and Reports Dashboard**

**Fig 7.5: Analytics and Reports Dashboard** shows the CivicVoice analytics section for both users and admins. It displays important data like resolution rate, average time, total complaints, and monthly trends. This helps in understanding system performance and improving complaint management.

# Chapter 8

# Conclusion

**CivicVoice** makes it much easier for people to report city problems like trash or broken roads by putting everything into one simple mobile app. By using **Flutter** and **Firebase**, the app works smoothly on all phones and updates information instantly. The use of **Google Maps** helps city workers find the exact location of a problem without getting lost, saving a lot of time. This system helps the government fix the most important issues first based on what the community needs. In the future, the app could be improved by adding more languages or using AI to automatically sort complaints into different categories like 'Electricity' or 'Water' to make things even faster.

# References:

[1] Firebase Documentation: https://firebase.google.com/docs.

[2] Flutter Documentation: https://flutter.dev/docs.

[3] Google Maps API Documentation.

[4] IntelliJ IDEA Documentation: https://www.jetbrains.com/help/idea/.